

llmXive Automated Review of In-Context World Modeling for Robotic Control

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a compelling method for in-context world modeling, but several specific quantitative claims in the text do not align with the reported data in the tables or the stated experimental protocols.

First, in Section 4.1, the total episode count is calculated as “ $500 \times 15 \times 4$ ”. The text explicitly defines the viewpoint split as 8 in-domain and 6 out-of-domain (total 14). The number 15 is inconsistent with the described protocol ($8+6=14$), leading to a discrepancy in the total episode count (28,000 vs. 30,000).

Second, the real-world results in Section 4.2 cite a drop from “68% to 17%”. The provided tables (Tab 1 and Tab 2) show the Multi-View (MV) baseline averages ranging from 30.8% to 74.5% in-domain and 19.8% to 48.3% out-of-domain. Neither 68% nor 17% appears as a direct average in the tables. These numbers likely refer to a specific task or a different aggregation not explicitly defined in the text, creating a mismatch between the narrative and the evidence.

Third, the analysis of false context in Section 5.1 contains a mathematical inconsistency. The text states the gain from correct context is “+13.6%”, citing the difference between ICWM (25.0) and “w/o ctx” (22.0). However, $25.0 - 22.0 = 3.0$, which is a 13.6% *relative* increase over 22.0, but the text frames it as a magnitude symmetric to the drop from “w/o act” (21.6) to ICWM (25.0), which is a 3.4 point drop (15.7% relative). The phrasing “symmetric in magnitude” is confusing and the percentage cited (13.6%) seems to be a relative calculation that doesn’t match the “magnitude” comparison intended.

Finally, the morphological generalization claim in Section 5.3 of a “stable margin of +60%” is not supported by the aggregate data (25.0 vs 19.8 is ~26% relative) nor the specific 80mm case (14.4 vs 5.6 is ~157% relative). The claim requires clarification on which specific metric or condition yields the 60% figure.

These are primarily issues of precision and alignment between the narrative and the data tables, which can be resolved by correcting the numbers or clarifying the specific subsets they refer to.

Action items.

- **[writing]** Section 4.1 claims ‘ $500 \times 15 \times 4$ total episodes’ for the LIBERO benchmark. However, the text specifies 4 suites, 8 training + 6 OOD viewpoints (14 total), and 500 episodes per viewpoint. The calculation $500 \times 14 \times 4 = 28,000$, not 30,000. The ‘15’ figure is unsupported by the stated protocol ($8+6=14$). Correct the total count or clarify the viewpoint split.
- **[writing]** Section 4.2 states ‘standard VLA performance drops sharply from 68% to 17%’. Table 1 (Seen) shows the MV baseline average is 74.5% (Spatial) to 30.8% (Long), and Table 2 (Unseen) shows 48.3% to 19.8%. Neither the 68% (seen) nor 17% (unseen) figures match the reported averages in the tables. Verify the specific task or suite these numbers refer to, or correct them to match the table averages.
- **[writing]** Section 5.1 claims ‘false context performs worse than no context at all (18.9 vs. 22.0)’. Table 3 confirms these averages. However, the text claims this negative transfer is ‘symmetric in magnitude to the gains from correct context (+13.6%)’. The gain from ‘w/o

ctx' (22.0) to 'ICWM' (25.0) is +3.0, not +13.6. The 13.6% figure appears to be the drop from ICWM to 'w/o act' (25.0 -> 21.6), not the gain over 'w/o ctx'. Correct the comparison or the percentage cited.

- **[writing]** Section 5.3 claims 'ICWM consistently outperforms MV across all offsets with a stable margin of +60%'. Table 2 (Unseen) shows MV at 19.8% and ICWM at 25.0% (avg), a margin of ~5.2 points (26% relative). The text likely refers to the specific 80mm spacer case (MV 5.6 vs ICWM 14.4, which is ~157% relative increase, not 60%). The 'stable margin of +60%' claim is not supported by the aggregate data or the specific 80mm case. Clarify the specific condition or correct the percentage.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively illustrates the conceptual framework of ICWM and the training pipeline, but the embedded quantitative charts lack necessary axis labels and definitions to fully support the visualized performance claims.

Figure 2

Figure 2 provides a clear and well-structured overview of the ICWM training and inference pipeline. The diagram effectively visualizes the flow from self-exploration to the Pretrained VLM, with distinct sections for training and inference that align perfectly with the provided caption.

Figure 3

The figure illustrates the task suite but is visually incomplete due to white blocks obscuring the bottom section. Additionally, the caption's claim of six distinct viewpoints is not visually supported by the provided panels.

Figure 4

The figure is visually clear but contains a likely data labeling error in the 'LONG' subplot where unseen performance exceeds seen performance, and missing baseline data points for the 'UNSEEN' group in that same subplot.

Figure 5

The figure effectively illustrates the qualitative differences between the baseline and the proposed method, but the caption’s use of (a)/(b) does not match the figure’s numeric labels (1/2), and there is unexplained text (‘Base:’) in the empty space below.

Figure 6

The figure displays t-SNE visualizations but lacks a caption entirely, preventing verification of the data source or the meaning of the labels. Additionally, the differing axis scales between the two subplots could be misleading without further explanation.

Figure 7

Figure 7 presents bar charts comparing methods under different conditions but fails to label axes or units, and the caption is insufficient to explain the semantic perturbations or the specific metrics being evaluated.

Figure 8

The bar chart clearly displays inference times for different shot settings, but the caption is grammatically incorrect and lacks necessary experimental context, while the axis labeling is slightly ambiguous regarding units.

Figure 9

Figure 9 effectively visualizes four distinct robotic manipulation tasks in simulation, with clear panel labels and descriptive titles for each sequence. The images are legible, and the caption accurately reflects the content shown, supporting the claim of successful multi-stage execution.

Figure 10

The figure effectively visualizes the morphological changes, but the caption text is inconsistent with the image labels regarding the specific percentage values and their order, and contains a minor LaTeX formatting artifact.

Figure 11

The figure displays successful real-robot trajectories for three tasks, but the claim of demonstrating robustness against object disturbances is unsupported by the visual evidence, which shows no perturbations.

Figure 12

The figure displays a grid of robot interaction pairs but is rendered useless by a placeholder caption (‘Enter Caption’) that fails to explain the content or the meaning of the labels.

Action items.

- **[writing]** Figure 1: The ‘OOD Viewpoint Shifts’ bar chart lacks a Y-axis label defining the metric (e.g., Success Rate), making the quantitative claims ambiguous.
- **[writing]** Figure 1: The ‘Real World Experiments’ panel contains a text label ‘6 OOD Viewpoint’ that is not defined in the caption or legend.
- **[science]** Figure 3: The caption claims the tasks are ‘evaluated across six distinct camera viewpoints,’ but the figure only displays a ‘Multi-Cam View’ and a ‘Single Camera View’ without showing or labeling the six specific viewpoints mentioned.
- **[writing]** Figure 3: The bottom section of the figure is partially obscured by large white blocks, cutting off the ‘Single Camera View’ label and the associated image sequence, making the figure incomplete.
- **[science]** Figure 4: The ‘LONG’ subplot shows ‘UNSEEN’ performance for MV, EXP, and ICWM that is higher than their ‘SEEN’ performance, which contradicts the definition of ‘Unseen (OOD)’ as a harder domain; verify if the x-axis labels are swapped or if the data is mislabeled.
- **[writing]** Figure 4: The ‘LONG’ subplot’s ‘UNSEEN’ group lacks the small baseline bars (pi-FAST, pi-0.5, NORA) visible in other subplots, making it unclear if they are zero or missing data.
- **[writing]** Figure 5: The caption references items (a) and (b), but the figure uses numeric labels ‘1’ and ‘2’ instead; update the caption to match the figure or vice versa.
- **[writing]** Figure 5: The text ‘Base:’ appears in the lower-left white space without any corresponding image or explanation, creating confusion.
- **[fatal]** Figure 6: The figure has no caption provided (‘(no caption)’), making it impossible to verify what the t-SNE plots represent or what the numerical labels (e.g., 45, 135) signify.
- **[science]** Figure 6: The x-axis scales differ between the ‘Spatial’ (-75 to 75) and ‘Object’ (-100 to 100) plots, which may mislead readers regarding the relative spread of the clusters without explicit normalization context.
- **[writing]** Figure 7: The caption ‘Semantic perturbations’ is too vague and does not describe the specific experiments, metrics, or conditions shown in the bar charts.
- **[science]** Figure 7: The bar charts lack axis labels and units, making it impossible to determine what metric is being measured (e.g., success rate, error) or the scale of the values.
- **[science]** Figure 7: The ‘Morphology’ section shows images of a robot with ‘adding rigid spacers’ but the corresponding chart labels refer to ‘ ΔL ’ (length change), creating a disconnect between the visual modification and the quantitative data.
- **[writing]** Figure 8: The caption contains a grammatical error (‘different setting’ should be ‘different settings’) and lacks specific details about the experimental conditions (e.g., hardware, model size) shown in the chart.

- **[science]** Figure 8: The x-axis label ‘Inference Time Cost Per-step’ is ambiguous; it is unclear if the values represent wall-clock time or computational cost, and the unit ‘s’ (seconds) is only shown inside the bars rather than on the axis.
- **[writing]** Figure 10: The caption lists percentages as ‘100%, 90%, 80%, 70%’, but the image labels show ‘Original Length’, ‘70% Length’, ‘90% Length’, and ‘80% Length’. The order and inclusion of ‘Original’ (vs 100%) are inconsistent between the text and the visual labels.
- **[writing]** Figure 10: The caption contains a formatting artifact ‘\100\%’ which should be corrected to ‘100%’ for proper typesetting.
- **[science]** Figure 11: The caption claims to illustrate ‘robustness against object disturbances,’ but the visualized trajectories show clean, successful executions without any visible perturbations or disturbances to the objects.
- **[writing]** Figure 11: The image contains three distinct task sequences (lifting, pick-and-place, stacking) labeled 1, 2, and 3, but the caption does not explicitly map these numbers to the specific tasks shown.
- **[fatal]** Figure 12: The caption is a placeholder (‘Enter Caption’) and does not describe the content, making the figure’s purpose and the meaning of the labels (o^s , o^e) impossible to interpret.
- **[science]** Figure 12: The image consists of a grid of robot interaction pairs labeled o^s and o^e without any explanatory text or context, failing to communicate the scientific claim or comparison being illustrated.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper demonstrates a high standard of accessibility for a competent reader from an adjacent field (e.g., a robotics or machine learning PhD). The authors consistently define acronyms and notation at their first occurrence, ensuring that the specialized vocabulary of In-Context World Modeling (ICWM) does not create barriers to comprehension.

Specifically, the term “Vision-Language-Action (VLA)” is expanded in the abstract and again in the Introduction. The core acronym “In-Context World Modeling (ICWM)” is defined immediately upon introduction in the abstract and the Introduction. The symbol ψ representing “system configuration” is explicitly defined in the Introduction and rigorously formalized in Section 3 (“Preliminary and Motivation”) before being used in equations. Similarly, the interaction context $\mathcal{T}$ and the inference function $\Psi(\mathcal{T})$ are clearly defined in Section 3.2 and 4.1 respectively, with their roles in the equations explained in the surrounding prose.

The paper avoids undefined in-group shorthand. While it references specific benchmarks

(LIBERO) and models (Qwen2.5-VL, FAST), these are standard in the field or accompanied by brief contextual descriptions (e.g., “action tokenizer” for FAST). The mathematical notation, including the POMDP formulation and mutual information terms, is introduced with clear definitions of the variables (s_k , ξ_k , o_k , a_k) and the assumptions (A1, A2) required for the propositions.

There are no instances of overloaded symbols or “obviously” statements that skip necessary definitions for an adjacent-field expert. The flow from problem formulation to method and experiments is self-contained regarding terminology. No action items are required.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent argument for In-Context World Modeling (ICWM) as a solution to system identification in VLA models. The logical flow from the problem definition to the proposed method and experimental validation is generally sound. However, there are specific inconsistencies between the numerical claims in the text and the data presented in the tables, which break the internal consistency of the argument.

In Section 4.2 (“Simulation Results”), the text states that ICWM improves the OOD success rate by **13.0%** over the Multi-View BC (MV) baseline. However, Table 2 reports the average OOD success rate for ICWM as **25.0%** and for MV as **19.8%**, a difference of only **5.2%**. Similarly, the text claims a **9.5%** improvement over the Explicit Configuration (EXP) baseline, while Table 2 shows a difference of **4.8%** (25.0% vs 20.2%). These discrepancies suggest the text may be citing results from a specific task suite or viewpoint without qualification, creating a non-entailed conclusion where the stated statistic does not follow from the aggregate data provided in the same section.

Additionally, in Section 4.3, the text argues that the negative transfer from “false context” is symmetric to the gains from correct context, citing a **+13.6%** gain. Table 1 shows the actual gain from adding context (ICWM 25.0% vs. “w/o context” 22.0%) is **3.0%**, and the loss from false context is **3.1%**. The **13.6%** figure cited in the text actually corresponds to the performance drop when removing *actions* (25.0% vs. 21.6%), not the gain from context. This is a clear mismatch between the textual claim and the supporting table data, undermining the precision of the argument regarding the symmetry of the effect.

These issues are classified as “writing” severity because they can be resolved by correcting the numbers in the text to match the tables or by clarifying that the percentages refer to specific subsets of the data. The core logical argument remains valid, but the quantitative support requires alignment to ensure the conclusions strictly follow from the presented premises.

Action items.

- **[writing]** Section 4.2 claims a 13.0% OOD improvement over MV, but Table 2 shows 25.0%

vs 19.8% (5.2% diff). Correct the text to match the table or specify the subset.

- **[writing]** Section 4.2 claims a 9.5% OOD improvement over EXP, but Table 2 shows 25.0% vs 20.2% (4.8% diff). Align the text with the reported aggregate data.
- **[writing]** Section 4.3 claims false context loss is symmetric to a +13.6% gain, but Table 1 shows the gain (ICWM vs w/o ctx) is only 3.0%. The 13.6% figure matches the ‘w/o actions’ drop. Correct the symmetry claim.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a strong core contribution regarding viewpoint generalization, supported by extensive simulation and real-robot experiments. However, the rhetoric in the Abstract, Introduction, and Conclusion occasionally overextends the scope of the demonstrated results to include semantic and morphological generalization as if they were equally robust findings.

Specifically, the Abstract and Introduction frame the method as solving generalization to “novel setups” and “robot morphologies” broadly. While the paper does include experiments on morphological changes (spacer attachments, link length scaling) and semantic variations (distractors, textures), the results in these areas are significantly more modest and context-specific than the viewpoint results. For instance, the morphological tests are limited to specific, controlled perturbations (e.g., rigid spacers of 20-80mm, specific link length ratios on a WindowX platform), and the semantic gains are described as “moderate.”

The current framing risks implying that the method has achieved broad, robust generalization across all these domains, when the evidence primarily supports a breakthrough in viewpoint adaptation, with promising but preliminary results in other areas. The conclusion should more carefully distinguish between the primary, well-supported finding (viewpoint generalization) and the secondary, more limited findings (semantic/morphological), perhaps by using qualifiers like “preliminary evidence” or “specific cases” for the latter. This would better align the paper’s claims with the actual strength and scope of the evidence provided.

Action items.

- **[writing]** The paper presents a strong core contribution regarding viewpoint generalization, supported by extensive simulation and real-robot experiments. However, the rhetoric in the Abstract, Introduction, and Conclusion occasionally overextends the scope of the demonstrated results to include semantic and morphological generalization as if they were equally robust findings. Specifically, the Abstract and Introduction frame the method as solving generalization to “novel setups” and “robot morphologies” b

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This work presents a method for In-Context World Modeling (ICWM) to improve robot generalization via test-time self-exploration. The research involves standard robotic manipulation tasks (stacking, lifting, pick-and-place) in simulation (LIBERO) and on a physical UR5e platform.

From a safety and ethics perspective, the paper is low-risk. The “active probing” phase described in Section 4.2 and Appendix B involves the robot executing random, task-agnostic movements within a pre-defined, kinematically constrained workspace to infer camera viewpoints and morphology. The authors explicitly state that this workspace is derived from the robot’s joint limits and is designed to “avoid contact with task-relevant objects” and ensure motions remain within “operable joint limits.” This is a standard safety procedure in robotics research (often called “calibration” or “self-probing”) and does not constitute a dual-use capability for harm, surveillance, or deception.

There are no human-subjects data, PII, or sensitive datasets involved; the human demonstrations mentioned (Appendix B) are standard teleoperation logs for training, and the paper does not release raw video of humans or identifiable data. The method does not lower the barrier for cyber-attacks or biological hazards. No specific, non-trivial risks requiring mitigation or disclosure were identified in the text. The work adheres to standard norms for safe robotic experimentation.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling hypothesis: that self-generated interaction context can serve as a system identifier for VLA models. The experimental design is generally sound in its use of controlled baselines (MV, EXP) and ablation studies (w/o actions, w/o images). However, the evidentiary strength of the central quantitative claims is currently undermined by a lack of reported variance and potential confounds in the training setup.

First, the headline result in Table 2 (25.0% vs 19.8% OOD success) is presented as a single aggregate number. In robotic manipulation benchmarks, performance can fluctuate significantly based on random seeds, initialization, and the specific distribution of test episodes. Without reporting the standard deviation across multiple seeds (e.g., 3-5 runs), it is impossible to determine if the 5.2% absolute gain is a robust effect or within the noise floor of the evaluation protocol. The “false context” ablation is a strong qualitative control, but it does not substitute

for statistical rigor on the primary metric.

Second, there is a potential confound in the comparison between ICWM and the Multi-View (MV) baseline. The ICWM model is trained with $N = 5$ interaction clips prepended to every sample, effectively increasing the context window length and the computational cost per training step compared to the MV baseline, which processes only the task query. While the authors argue the gain comes from the *content* of the context, the design does not rule out that the gain comes from the *capacity* to attend to a longer sequence or the implicit regularization of processing more tokens. A fair comparison requires a baseline that also processes $N = 5$ tokens (e.g., dummy tokens or repeated observations) to ensure the “context” benefit is not merely a “longer context” benefit.

Finally, the real-world results (Section 4.3) rely on a single aggregate success rate across 600 trials. While the sample size is decent, the lack of per-viewpoint variance reporting makes it difficult to assess stability. If the baseline failed catastrophically on just two of the six viewpoints while performing well on the others, the aggregate drop might be misleading. Reporting the mean and standard deviation across the 6 viewpoints (or the 6 seeds if multiple seeds were used) would provide a clearer picture of the method’s reliability.

Addressing these points—specifically by adding seed variance to the main tables and clarifying the compute/context length control—would significantly strengthen the claim that the observed improvements are due to the proposed mechanism rather than experimental noise or confounding factors.

Action items.

- **[writing]** The paper presents a compelling hypothesis: that self-generated interaction context can serve as a system identifier for VLA models. The experimental design is generally sound in its use of controlled baselines (MV, EXP) and ablation studies (w/o actions, w/o images). However, the evidentiary strength of the central quantitative claims is currently undermined by a lack of reported variance and potential confounds in the training setup. First, the headline result in Table 2 (25.0% vs 19.8% OOD su

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical reporting in this paper is currently insufficient to support the quantitative claims of “significant” improvement and “consistent” gains. While the experimental design appears sound, the analysis of the resulting data lacks necessary inferential machinery.

Specifically, Tables 1, 2, and 3 report point estimates (success rates) to one decimal place but omit any measure of dispersion (standard deviation, standard error, or confidence intervals). In deep learning experiments involving stochastic training, a single number is not a reliable estimator of true performance. The text mentions “25 trials per task,” but it is unclear if reported

averages aggregate over multiple independent training seeds. If results are from a single seed, the reported precision is misleading; if aggregated, variance across seeds must be reported to assess stability.

Furthermore, the authors repeatedly use the term “significantly” (e.g., Abstract, Section 4.2) to describe performance gaps (e.g., +13.0%). In scientific writing, this implies a statistical hypothesis test was conducted and the null hypothesis rejected. No such tests are described, nor are p-values or CIs provided. Without this, the claim of “significance” is unsupported.

Finally, the paper presents numerous pairwise comparisons across viewpoints and ablation conditions, highlighting specific cells as “best” or “worst” without addressing the multiple comparisons problem. If statistical tests are added, a correction (e.g., Holm-Bonferroni) is required to avoid inflating the Type I error rate.

These are reporting and analysis gaps that can be fixed by re-running analysis on existing data or adjusting language to match the evidence provided.

Action items.

- **[writing]** Tables 1-3 report success rates to one decimal place but omit uncertainty metrics (SD/SE/CI). Given stochastic training, report mean $\pm$ SD across 3 seeds or state if single-seed results are used.
- **[writing]** Sections 4.2-4.3 claim ‘significant’ improvements without reporting p-values, CIs, or hypothesis tests. Either report test statistics/p-values for key comparisons or rephrase to ‘higher average’ without invoking significance.
- **[writing]** Tables 2-3 present multiple pairwise comparisons across viewpoints/ablations without multiplicity correction. If adding tests, apply Holm-Bonferroni or Benjamini-Hochberg correction to control false discovery rate.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and the core argument flows logically from the problem statement to the proposed solution and validation. However, there are specific instances where sentence construction and paragraph organization impede the reader’s momentum.

In Section 4.2 (Real-world Results), the sentence beginning “As qualitatively shown in \autoref{fig:case}, while the base VLA exhibits...” is overly long and complex. It stacks a dependent clause, a contrastive clause, and a main clause in a way that forces the reader to hold too much information before reaching the verb “utilizes.” Splitting this into two sentences—one describing the baseline failure and one describing the ICWM success—would significantly improve clarity.

In Section 4.3 (Analysis), the paragraph introducing the ablation study lists five settings

but immediately pivots to interpreting the table results without a clear bridging sentence. The reader is left to infer the connection between the setup description and the data presentation. A simple transition sentence explicitly stating that the table follows and what it reveals would smooth this handoff.

Additionally, in Section 4.4, the textual enumeration of probing strategies (a-d) does not perfectly align with the visual order of columns in Table 2, creating a minor cognitive load for the reader trying to map the text to the data. Ensuring the text order matches the table order, or explicitly cross-referencing them, would eliminate this friction.

Finally, in Appendix A.2, a figure is inserted in the middle of a numbered list describing experimental tasks. This breaks the visual and logical continuity of the list. Moving the figure to the end of the list or restructuring the text to accommodate the figure would restore the paragraph’s coherence.

These issues are minor and fixable through careful editing, but they currently require the reader to pause and re-parse specific sections.

Action items.

- **[writing]** The paper is generally well-structured and the core argument flows logically from the problem statement to the proposed solution and validation. However, there are specific instances where sentence construction and paragraph organization impede the reader’s momentum. In Section 4.2 (Real-world Results), the sentence beginning “As qualitatively shown in \autoref{fig:case}, while the base VLA exhibits...” is overly long and complex. It stacks a dependent clause, a contrastive clause, and a main cl